1. Problem Definition

The increase in usage of low-cost unmanned aerial systems has created many airspace security problems, as seen in the 2018 Gatwick Airport shutdown and in the widespread commercial quadcopter use in the ongoing Ukraine conflict. Passive Acoustic Sensing is complementary to Radar and Radio-Frequency Detectors since passive acoustic sensors are inexpensive, operate covertly, and do not require line-of-sight or radio-frequency emissions from the target. However, real world use has three requirements that raw accuracy alone does not capture: generalization to new and unknown recording conditions, discrimination between drones and legitimate rotorcraft such as medical helicopters, tolerance of low signal-to-noise ratio (SNR).

This project will be a three-class classification problem: drone, helicopter, background, where the helicopter class is included to test whether models can distinguish target drones from acoustically similar legitimate rotorcraft rather than dismissing them both into a single aircraft category.

2. Dataset Description

This project uses three public data sets. The Svanström Multi-Sensor Drone Detection dataset (Svanström et al., 2021) is the primary dataset and contains drone, helicopter and background recordings from three different Swedish airports. The Al-Emadi DroneAudioDataset (Al-Emadi et al., 2019) which includes additional drone audio to support the drone class and provides alternative sources for the cross-source generalisation evaluation. The ESC-50 (Piczak, 2015) dataset contains environmental noise to degrade the signal-to-noise ratio during testing; the Svanström set contains 90 ten-second recordings (30 per class) at 44.1 kHz; segmentation into 2-second sliding windows with 50% overlap giving ~810 labelled examples, resampled to 32 kHz mono to match the PANNs input specification. Class balance is somewhat uniform across drones, helicopters and backgrounds by design. Al-Emadi is reserved in full as the cross-source evaluation set and does not enter training. To avoid leakage between train/validation/test splits that could occur when using a small number of labeled audio examples, the data is split at the recording level, the audio is converted to 32 kHz mono to match the PANNs input specification, and then cached as log-mel spectrograms on Google Drive to keep per-epoch Colab memory and compute low.

3. Proposed Approaches

Three architectures are compared, which spans across classical margin-based classification, convolutional learning to transfer learning. Drones produce a characteristic harmonic signature from the propellers frequencies which differs from helicopter and other environmental signatures (Al-Emadi et al., 2019); each model captures this differently. The first architecture utilizes a linear Support Vector Machine (SVM) over Mel Frequency Cepstral Coefficients (MFCC), Spectral Centroid, Roll Off, Band Width, Zero Crossing Rate and Root Mean Square Energy, providing a baseline performance based upon hand engineered features. Architecture 2 is a five-block CNN trained from scratch on 128-band log-mel spectrograms, following the spectrogram-as-image paradigm of Hershey et al. (2017); Casabianca and Zhang (2021) reported 94.7% mean accuracy on a comparable drone task using this approach. Architecture 3 uses PANNs CNN10 pretrained on AudioSet (Kong et al., 2020). PANNs are preferred over ImageNet-pretrained vision networks because they match the audio domain, placing this task in the small-data, similar-domain quadrant where fine-tuning is the indicated strategy: fine-tuned PANNs achieved 0.947 accuracy on ESC-50 while the same network trained from scratch achieved 0.833 (Kong et al., 2020).

Fine-tuning proceeds in three stages of progressive unfreezing, an approach rarely explored in prior drone detection work: only the head is trained at learning rate 1x10e-3, the final convolutional block is unfrozen with differential learning rates (1x10-5 for convolutional layers, 1x10e-3 for the head), finally the final two blocks are unfrozen on the same schedule. Macro F1 and per class AUC ROC are the primary metrics for measuring performance and are calculated with bootstrap 95% confidence interval (1000 resamples). Three aspects will be considered when evaluating performance: Stratified in distribution testing, a cross-source generalization gap measured via held out source data, and an SNR degradation curve at clean, +10 dB, +5 dB, 0 dB, and -5 dB.

4. Challenges

The limited number of samples is mitigated through the use of SpecAugment time and frequency masks (Park et al., 2019) and MixUp augmentation, both of which follow the same pipeline that was used to train PANNs (Kong et al., 2020). Leakage between windowed segments is prevented through the recording level split. The potential loss in performance due to a cross-source domain shift shown by Al-Emadi et al. (2019) to significantly decrease performance on unseen drone types is evaluated as part of the research question and not hidden. Over-fitting to the relatively small training set is handled through Dropout (0.3 to 0.5), L2 Weight Decay (1x10-4), and Early Stopping based on Macro F1 Validation Performance. An alternative from fine tuning PANNs to feature extraction from frozen features is also available, where it achieved an accuracy of .918 on ESC-50 (Kong et al., 2020).

5. Timeline

Four weeks of active development will take place before the project deadline of April 24th, 2026. In week one (March 30-April 5) the datasets will be collected and preprocessed along with creating the recording level split manifests. In week two (April 6-12) the classical baseline model and CNN model will be trained and validated. In week three (April 13-19) three-stage PANNs fine-tuning, cross-source holdouts, and SNR curves with bootstrap intervals will occur. In week four (April 20-24) the draft is completed, figures created, and checks for reproducing results with restart kernel will occur, along with time to debug.